

## COURSE SYLLABUS

### CSC00004 – Introduction to Information Technology

#### 1. GENERAL INFORMATION

|                              |                                        |
|------------------------------|----------------------------------------|
| Course name:                 | Introduction to Information Technology |
| Course name (in Vietnamese): | Nhập môn Công nghệ Thông tin           |
| Course ID:                   | CSC00004                               |
| Knowledge block:             | Fundamental Education                  |
| Number of credits:           | 4                                      |
| Credit hours for theory:     | 45                                     |
| Credit hours for practice:   | 30                                     |
| Credit hours for self-study: | 90                                     |
| Prerequisite:                | None                                   |
| Prior-course:                | None                                   |

#### 2. COURSE DESCRIPTION

The course will give students an overview of the history of the development and formation of the Information Technology (IT). In addition, general knowledge includes the basic knowledge about data storage and manipulation, numeral systems, computer architecture, operating systems, computer networks, and computer safety are introduced. Students are provided an overview of the role of software development, database and information systems, as well as some recent popular trends such as artificial intelligence, natural language processing, image processing, etc. are included. The course also helps students orientating their study and career development through the introduction of majors and their related courses in the IT domain. Furthermore, it provides students with the skills (such as teamwork, critical thinking, problem solving and creative thinking) and professional ethics necessary to work and communicate in the learning and professional environments later.

### 3. COURSE GOALS

At the end of the course, students are able to

| ID | Description                                                                                                                                                                                                                |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G1 | Work at an individual and team level to implement and present some topics related to the general knowledge, career orientation. Demonstrate critical thinking and creative thinking in the process of implementing topics. |
| G2 | Present the meaning, role, current status of hardware devices, software. Apply software, technology platforms to collaborate, present, share and backup data.                                                              |
| G3 | Explain the data storage platform in a computer storage and describe the roles of database and database administration system.                                                                                             |
| G4 | Identify the role of the operating system, the development process, and distinguish different types of operating systems.                                                                                                  |
| G5 | Present basic knowledge about computer networks, internet, network protocols, devices, connection issues and security issues.                                                                                              |
| G6 | Summarize the main characteristics of artificial intelligence and its related fields and describe some emerging technologies.                                                                                              |
| G7 | List some important rules and regulations. Explain some characteristics of professional ethics.                                                                                                                            |

### 4. COURSE OUTCOMES

| CO   | Description                                    | I/T/U |
|------|------------------------------------------------|-------|
| G1.1 | Organize, operate and manage team in teamwork. | T     |

|      |                                                                                                                           |   |
|------|---------------------------------------------------------------------------------------------------------------------------|---|
| G1.2 | Propose ideas, analysis and discuss these ideas.                                                                          | T |
| G1.3 | Present the topic in class and make a report.                                                                             | T |
| G2.1 | Present the meaning, role, current status of hardware devices, software.                                                  | T |
| G2.2 | Apply software, technology platforms to collaborate, present, share and backup data.                                      | T |
| G3.1 | Explain the data storage platform in a computer.                                                                          | T |
| G3.2 | Describe roles of database and database administration system                                                             | T |
| G4.1 | Identify the role of the operating system, the development process, and distinguish different types of operating systems. | T |
| G5.1 | Present basic knowledge about computer networks, internet, network protocols.                                             | T |
| G5.2 | List devices, connection issues and security issues.                                                                      | T |
| G6.1 | Summarize the main characteristics of artificial intelligence and related fields.                                         | T |
| G6.2 | Describe some emerging technologies.                                                                                      | T |
| G7.1 | List some important rules and regulations.                                                                                | I |
| G7.2 | Explain some characteristics of professional ethics.                                                                      | T |

## 5. TEACHING PLAN

| ID | Topic | Course outcomes | Teaching/Learning Activities (samples) |
|----|-------|-----------------|----------------------------------------|
|----|-------|-----------------|----------------------------------------|

|   |                                                                                                                                                                                                                                        |            |                                                   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------|
| 1 | <p>Introduce an overview of the development process, the role of IT in life.</p> <p>Learn about computer classification, the basic architecture of a computer. Introduce the characteristics and role of software in the computer.</p> | G2.1       | <p>Lecturing</p> <p>Group discussion</p>          |
| 2 | <p>Introduce basic hardware devices such as ports, circuits to peripheral devices such as RAM, hard drives, ... in computers. Identify the characteristics, roles and meanings of each device and how they work together.</p>          | G2.1       | <p>Lecturing</p> <p>Demonstration</p>             |
| 3 | <p>Principle of storing information in computers such as number (integer and floating point), text, images, sounds, ...</p> <p>Basic operations on the counting system.</p>                                                            | G2.2       | <p>Lecturing</p> <p>Demonstration, discussion</p> |
| 4 | <p>How the CPU executes a low-level program. How the CPU works with other devices to make a request.</p> <p>The relationship between low-level languages and high-level programming languages.</p>                                     | G3.1       | <p>Lecturing</p> <p>Quiz</p>                      |
| 5 | <p>Operating system (OS): characteristics, roles, and development process on all computer platforms. The basic architecture and components of the operating system.</p>                                                                | G4.1       | <p>Lecturing</p>                                  |
| 6 | <p>Computer network: transmission principles, connection protocols, network</p>                                                                                                                                                        | G5.1, G5.2 | <p>Lecturing</p>                                  |

|    |                                                                                                                                                                                                                                                      |                  |                             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------|
|    | types, security. Supported devices and software.<br><br>Computer security such as how to attack and how to protect the computer.<br><br>Introduction to the major of Computer networks at FIT-HCMUS                                                  |                  | Discussion                  |
| 7  | Database system: roles, meanings, applications. Database management systems.<br><br>Introduction to the major of Information Systems at FIT-HCMUS                                                                                                    | G3.2             | Lecturing<br><br>Discussion |
| 8  | Introduction to artificial intelligence, machine learning, and applications.<br><br>Introduction to the major of Computer Science at FIT-HCMUS                                                                                                       | G6.1             | Lecturing<br><br>Discussion |
| 9  | Software engineering: The Software Engineering Discipline, The Software Life Cycle, Development Phases, Software Engineering Methodologies, Documentation, Project Management.<br><br>Introduction to the major of Software Engineering at FIT-HCMUS | G6.2             | Lecturing<br><br>Discussion |
| 10 | Professional ethics such as copyright laws, privacy and legal documents related to IT. Rules and regulations in university and company.                                                                                                              | G7.1, G7.2, G7.3 | Case study, discussion      |
| 11 | Introduction to FIT-HCMUS<br>Introduction to IT certificates                                                                                                                                                                                         | G7.1, G7.2, G7.3 | Lecturing<br><br>Discussion |

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the

corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

## 6. ASSESSMENTS

| Topic                                | Description                                                       | Ratio (%)  |
|--------------------------------------|-------------------------------------------------------------------|------------|
| <b>Assignments at Class Homework</b> | The content and number of exercises are specified by each teacher | <b>10%</b> |
| <b>Practice</b>                      | Weekly exercises and 1 or 2 projects                              | <b>40%</b> |
| <b>Final Exams</b>                   | Students are not allowed to use the material                      | <b>50%</b> |

**Bonus: maximum of 1 point based on class activities and assignments.**

## 7. RESOURCES

### Textbooks

- Computer science: reflection on the field , reflections from the field, Committee on the Fundamentals of Computer Science, National Academies Press, 2004.
- Computer science: an overview, Gleen Brookshear, Dennis Brylow, 12th edition, 2014
- Computers are your future, Introductory, Catherine Laberta, 12th Edition, 2011
- Ethics for the Information Age, Mike Quinn, 7th Edition, 2016.

## 8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than **3** theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded **0** for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.